

Lily Oglesby

loglesby@marshall.usc.edu | [linkedin.com/in/lilyoglesby](https://www.linkedin.com/in/lilyoglesby) | lilyoglesby.com | github.com/tygurllile | +1-951-203-0801

Summary

Graduate business research student with a background in data science and astrophysics. Interested in using statistical analysis, machine learning, and business analytics to understand complex data and support practical decisions. Experience building reproducible Python and SQL workflows, developing predictive models, analyzing large datasets, and communicating findings across scientific, public health, and business applications.

Skills

- **Programming and Data Science:** Python, SQL, Pandas, NumPy, SciPy, Scikit-Learn, TensorFlow/Keras, PyMC, Matplotlib, Seaborn, Jupyter
- **Business Analytics and Visualization:** Excel, Solver, SolverTable, KPI Analysis, Tableau (coursework), Spreadsheet Modeling, Data Visualization
- **Databases and Development:** PostgreSQL, MongoDB, Git, GitHub, Linux, Docker, VS Code, LaTeX
- **Analytical Methods:** Regression, Classification, Random Forests, CNNs, Anomaly Detection, Time-Series Analysis, Forecasting, Statistical Inference, Hypothesis Testing, A/B Testing, Bayesian Inference, Causal Inference, Cross-Validation, Feature Engineering, Model Evaluation, Optimization, Segmentation, Retention and Churn Analysis
- **Communication and Collaboration:** Technical communication, cross-functional collaboration, teaching and mentoring, presenting analytical findings to technical and nontechnical audiences

Education

- **University of Southern California | Marshall School of Business** Los Angeles, CA
Master of Science in Business Research Aug 2025 – Dec 2026
 - **Fall 2026 coursework:** Regression Analysis, SQL Databases, Spreadsheet Modeling, Data Analytics for the Games Industry, Deep Learning
- **University of California, Berkeley** Berkeley, CA
Bachelor of Arts: Double Major in Data Science and Astrophysics Aug 2021 – May 2025

Research Experience

- **Haqdarshak Social Protection Program Analytics Project** 2025
USC Graduate Research, Data Analysis, Program Analytics
 - Cleaned and reviewed program data and analyzed field-worker activity over time to identify operational patterns affecting outreach and access to government benefit and subsidy programs.
 - Combined analytical findings with stakeholder research to develop recommendations for improving outreach, service delivery, and connections with eligible individuals.
- **Space Sciences Laboratory, UC Berkeley | lilyoglesby.com/jgr-paper** Berkeley, CA
Undergraduate Researcher - Data Analysis Aug 2021 – May 2025
 - Built reusable Python pipelines for three years of satellite data, performing data validation, time-series preprocessing, statistical filtering, event comparisons, and visualization to identify patterns across atmospheric and space-weather measurements.
 - Produced reproducible analyses and figures supporting a [peer-reviewed publication](#) in the *Journal of Geophysical Research*.
- **Institute for Astronomy, University of Hawai'i at Mānoa | lilyoglesby.com/cnn-research** Honolulu, HI
Research Intern - Machine Learning Jun 2024 – Aug 2024
 - Applied anomaly detection to approximately 800,000 astronomical images to identify unusual observations within a large imaging dataset.
 - Trained and evaluated a convolutional neural network for multiclass classification and used GPU parallelization to accelerate preprocessing and model execution. [Report available online.](#)

Selected Data Science and Analytics Projects

- **Particulate Matter Health Impact and Forecasting** 2024
Python, Pandas, Scikit-Learn, Statistical Modeling, Causal Inference
 - Integrated and analyzed CDC and NOAA data to study relationships between PM2.5 exposure and asthma outcomes and predict pollution levels from weather and environmental variables.
 - Compared generalized linear and random forest models using feature engineering, model evaluation, and leakage-resistant temporal validation.
- **PostgreSQL vs. MongoDB Yelp Benchmark** 2024
PostgreSQL, MongoDB, SQL, Python, Jupyter, Docker
 - Designed equivalent analytical workloads across relational and document databases and compared data models, SQL queries, aggregation pipelines, indexes, execution plans, scan volume, and runtime performance.
- **Scientific Data Analytics Portfolio** 2024
Python, Pandas, NumPy, Scikit-Learn, PyMC, Data APIs, Astropy
 - Built reproducible workflows using satellite observations, stellar time-series data, telescope images, and astronomical spectra; applied database queries, feature engineering, regression, cross-validation, Bayesian estimation, high-dimensional preprocessing, and model evaluation.

Teaching and Mentoring Experience

- **USC, UC Berkeley, and University of Hawai'i** Los Angeles, Berkeley, and Honolulu
Instructor, Teaching Assistant, Reader, Grader, and Research Mentor Aug 2022 – Jul 2026
 - Taught and mentored high school, undergraduate, and graduate students through USC, UC Berkeley, and the Mauna Kea Scholars Program; led workshops and office hours on responsible AI, Python, data analysis, research design, graduate applications, and technical presentations.